

Week 6 Worksheet Solutions

Jacob Erlikhman

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Exercise 1. The hamiltonian for the harmonic oscillator is

$$H = \frac{1}{2m}[p^2 + (m\omega x)^2].$$

Show that

$$H = \hbar\omega(a^\dagger a + 1/2),$$

where

$$\begin{aligned}a^\dagger &= \alpha x + i\beta p \\ a &= \alpha x - i\beta p.\end{aligned}$$

That is, determine α and β .

We compute

$$a^\dagger a = \alpha^2 x^2 + \beta^2 p^2 + \hbar\alpha\beta.$$

From this we can read off that

$$\begin{aligned}|\alpha| &= \sqrt{\frac{m\omega}{2\hbar}} \\ |\beta| &= \frac{1}{\sqrt{2m\hbar\omega}}.\end{aligned}$$

To determine the signs, we can solve the equation

$$\hbar\alpha\beta + \frac{1}{2} = 0.$$

This has solution

$$\alpha\beta = -\frac{1}{2\hbar}.$$

If we want to keep α positive, then we set

$$\beta = -\frac{1}{\sqrt{2m\hbar\omega}}.$$

Exercise 2. A particle of mass m moves in the potential

$$V(x) = \begin{cases} +\infty, & x \leq 0 \\ \frac{1}{2}m\omega^2 x^2, & x > 0 \end{cases},$$

a harmonic well with a rigid wall slammed down at the origin. (This models, for example, a molecule vibrating against a hard surface, and the same equation governs the $\ell = 0$ radial motion of the three-dimensional oscillator.) Use the analytic solution of the full oscillator from lecture, whose normalized eigenstates are

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}, \quad \xi \equiv \sqrt{\frac{m\omega}{\hbar}} x,$$

with energies $E_n = (n + \frac{1}{2})\hbar\omega$ and Hermite polynomials are defined by: $H_0 = 1$ with a recurrence relation $H_{n+1}(x) = 2xH_n(x) - H'_n(x)$.

- a) Because $V = \infty$ for $x \leq 0$, the wave function must vanish there, in particular $\psi(0) = 0$. Argue that the allowed stationary states of the half-oscillator are precisely the eigenstates of the *full* oscillator that vanish at the origin, and identify which n these are. Hence write the allowed energies E_k ($k = 0, 1, 2, \dots$).

Hint: One approach is to use the result of Exercise 1 and the fact that $\psi_n(x) = (a^\dagger)^n \psi_0(x)$.

- b) Write the normalized ground-state wave function of the half-oscillator explicitly, in terms of ξ and H_1 . (Watch the normalization: the state now lives only on $x > 0$.)
- c) Find the most probable position of the particle in this ground state.
- d) The full oscillator has ground-state energy $\frac{1}{2}\hbar\omega$, but the half-oscillator's is $\frac{3}{2}\hbar\omega$. Explain physically why walling off $x < 0$ raises the zero-point energy.

- a) **First Solution:** For $x \leq 0$ the potential is infinite, so $\psi \equiv 0$ there, and continuity forces $\psi(0) = 0$. For $x > 0$ the Schrödinger equation is *identical* to the full oscillator's, so every half-oscillator solution coincides on $x > 0$ with a full-oscillator eigenfunction that vanishes at the origin. The full eigenstates have definite parity, $\psi_n(-x) = (-1)^n \psi_n(x)$: an even state has an antinode at $x = 0$ ($\psi_n(0) \neq 0$), while an *odd* state passes through zero, $\psi_n(0) = 0$. So only the odd states survive, $n = 1, 3, 5, \dots$. Writing $n = 2k + 1$,

$$E_k = (n + \frac{1}{2})\hbar\omega = (2k + \frac{3}{2})\hbar\omega, \quad k = 0, 1, 2, \dots,$$

that is $\frac{3}{2}\hbar\omega, \frac{7}{2}\hbar\omega, \frac{11}{2}\hbar\omega, \dots$, evenly spaced by $2\hbar\omega$ (half the levels of the full oscillator are gone).

Alternate Solution: We can instead use the Hermite polynomials. Using the recursion relation

$$H_{n+1}(x) = 2xH_n(x) - H'_n(x)$$

and the relation $H'_n(x) = 2nH_{n-1}(x)$, we get that

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x).$$

Plugging in $x = 0$, we find

$$H_{n+1}(0) = -2nH_{n-1}(0).$$

Thus, since $H_0 \equiv 1$ is even, we find that all H_n with n even will not vanish at 0, so they can't satisfy the boundary condition there. On the other hand, $H_1(x) = 2x$ vanishes at 0, so all H_n with n odd will vanish at 0 and will satisfy the boundary condition there. Thus, we find that the solutions to the half oscillator are precisely the odd states of the full oscillator.

- b) The ground state ($k = 0$) is the full oscillator's $n = 1$ state, restricted to $x > 0$ and renormalized. Since ψ_1 is odd, $\int_0^\infty |\psi_1|^2 dx = \frac{1}{2}$, so we multiply by $\sqrt{2}$. With $H_1(\xi) = 2\xi$, so that $\psi_1 = (m\omega/\pi\hbar)^{1/4} \sqrt{2} \xi e^{-\xi^2/2}$,

$$\psi_0^{\text{half}}(x) = \sqrt{2} \psi_1(x) = \boxed{2 \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \xi e^{-\xi^2/2}} \quad (x > 0), \quad \psi_0^{\text{half}} = 0 \quad (x \leq 0).$$

Check: $\int_0^\infty |\psi_0^{\text{half}}|^2 dx = 4 \sqrt{\frac{m\omega}{\pi\hbar}} \sqrt{\frac{\hbar}{m\omega}} \int_0^\infty \xi^2 e^{-\xi^2} d\xi = \frac{4}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{4} = 1.$

- c) The density is $|\psi_0^{\text{half}}|^2 \propto \xi^2 e^{-\xi^2}$. Maximizing, $\frac{d}{d\xi}(\xi^2 e^{-\xi^2}) = 2\xi(1 - \xi^2)e^{-\xi^2} = 0$ gives $\xi = 1$ (the root $\xi = 0$ sits at the wall, a minimum), so the most probable position is

$$\boxed{x_{\text{mp}} = \sqrt{\frac{\hbar}{m\omega}}}.$$

Unlike the full oscillator, whose ground state peaks at the center $x = 0$, the half-oscillator's ground state peaks away from the wall.

- d) The wall forbids $x < 0$, exactly where the full oscillator's even ground state had its peak. The surviving state must vanish at the origin, so it carries a node there and hence more curvature, which means more kinetic energy. Equivalently, squeezing the particle into half the space is extra confinement, and by the uncertainty principle extra confinement costs energy. The half-oscillator ground state is literally the full oscillator's first excited state, with energy $\frac{3}{2}\hbar\omega$.

Exercise 3. Bonus. Continue in the setup of Exercise 1.

- Show that $[a, a^\dagger] = aa^\dagger - a^\dagger a = 1$.
- Show that $[a^\dagger a, a] = -a$.
- If $N = a^\dagger a$ and $N\psi = \lambda\psi$ for some scalar λ and some normalizable ψ , show that $Na\psi = (\lambda - 1)a\psi$.
- Show that if ψ is normalized, then

$$\int |a\psi|^2 dx = \int \left(\psi^*(\alpha x)^2 \psi + \psi^*(\beta p)^2 \psi \right) dx - 1/2.$$

Hint: Use integration by parts, noting that $(|\psi|^2)' = (\psi^*)'\psi + \psi^*\psi'$.

- e) Show that there is a normalized solution ψ_0 to the TISE such that $a^\dagger a \psi_0 = 0$.

Hints: Use part (e) to show that $\int |a\psi|^2 dx = \int \psi^* N \psi dx$. Together with part (d), what does this tell you about repeated applications of a to ψ ?

- f) What is the energy of ψ_0 ?

- g) Show that $N = a^\dagger a$ is the **number operator**: If $\psi_n = (a^\dagger)^n \psi_0$ is the n^{th} stationary state, then

$$N\psi_n = n\psi_n.$$

- h) Show that if $N\psi_0 = 0$, then a complete set of normalized solutions to the equation $N\psi = \lambda\psi$ is given by $(a^\dagger)^n \psi_0$, so that $(a^\dagger)^n \psi_0$ are also a complete set of solutions to the TISE.

The solution to this exercise is in the Week 7 solutions.